

Technical Data Sheet

NebuLink Gateway NLG8400

Document Date: January, 2024

Document Version: 1.0

Copyright ©2023, System Level Solutions, Inc. (SLS) All rights reserved. SLS, an Embedded systems company, the stylized SLS logo, specific device designations, and all other words and logos that are identified as trademarks and/or service marks are, unless noted otherwise, the trademarks and service marks of SLS in India and other countries. All other products or service names are the property of their respective holders. SLS products are protected under numerous U.S. and foreign patents and pending applications, mask working rights, and copyrights. SLS reserves the right to make changes to any products and services at any time without notice. SLS assumes no responsibility or liability arising out of the application or use of any information, products, or service described herein except as expressly agreed to in writing by SLS. SLS customers are advised to obtain the latest version of specifications before relying on any published information and before orders for products or services.

ds_nlg8400_1.0

How to contact SLS?

For the most up-to-date information about SLS products, go to the SLS worldwide website at <https://www.slscorp.com>. For additional information about SLS products, consult the source shown below.

Information Type	E-mail
Product literature services, SLS literature services, Non-technical customer services, or Technical support.	sales@slscorp.com

Information of Nebulae

Nebulae® is a business vertical of SLS to provide Internet of Things (IoT) solutions. For the most up-to-date information about Nebulae products and solutions, go to Nebulae worldwide website at <https://www.nebulae.io> For additional information, contact at sales@nebulae.io

Preface

Privacy Information

Table below shows the revision history of NebuLink Gateway NLG8400 data sheet.

Revision History

Version	Date	Description of Changes
1.0	January, 2024	Initial Release

Typographic Conventions

This data sheet uses the typographic conventions as shown below:

Visual Cue	Meaning
Bold Type with Initial Capital letters	All headings and Sub headings Titles in a document are displayed in bold type with initial capital letters; Example: General Description, Features.
Bold Type with Italic Letters	All Definitions, Figure and Table Headings are displayed in Italics. Examples: <i>Figure 1-1. NebuLink Gateway NLG8400, Table 1-1. NebuLink Gateway NLG8400 Specifications.</i>
1., 2.	Numbered steps are used in a list of items, when the sequence of items is important. such as steps listed in procedure.
•	Bullets are used in a list of items when the sequence of items is not important.

THIS PAGE IS LEFT BLANK INTENTIONALLY.

Table of Contents

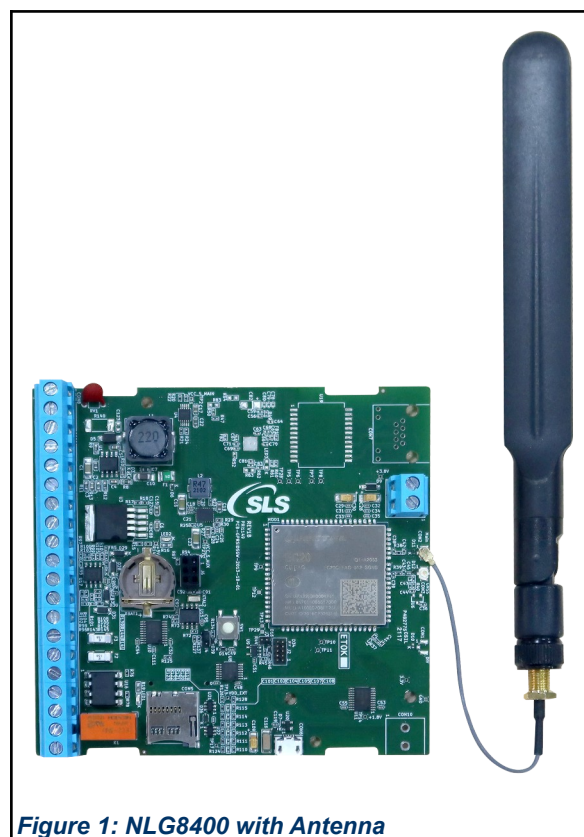
1	Introduction.....	8
2	Key Features.....	9
3	Application.....	10
4	Block Diagram.....	11
5	Technical Specifications.....	12
6	I/O Specifications.....	13
7	Package & Dimension.....	14
8	Pin Details.....	16

1 Introduction

NebuLink Gateway's industrial series close the gap of IoT implementation in industries 4.0. This product serves as SCADA RTU units for remote logic control and monitoring in various industrial applications. This offers an easy and user-friendly development support which allows users to implement control logic on controllers. With its powerful and easy-to-use features, it will minimize the learning curve, shorten time to market and reduce the labor and cost spent on system development.

NLG8400 is powered with ARM Cortex A7 processor with multiple interface such as RS232, RS485, I2C, Cellular to integrate in existing industrial and commercial environment and make them smart solution. Large external storage support using SD card, Analogue and Digital input and outputs, compact size, easy fitting and battery support makes it unique for any battery based M2M solutions for quick implementation. It supports QuecOpen™ SDK for programming and configuration with validated Modbus RTU, Modbus TCP and DLMS stack. This ease the integration with any SCADA software available.

With all these features, NLG8400 helps the end user to shorten their product cycle time to the market.



2 Key Features

The key features of the NLG8400 are:

- Built-in Lightweight Web Server for GSM and Peripherals
- Supports I/O, Counter, Timer
- Modbus/TCP Protocol for SCADA Software Seamless Integration
- RS-232 and Relay Interfaces
- 2-Way Isolation/ESD Protection
- Analogue and Digital IO support
- Micro SD of storage expansion for Data Logging and configuration recovery
- 3.7V/4.2V Rechargeable Lithium Ion/Polymer Battery Operated Solution to take on control on absence of Main supply
- RTC Supported Solution to achieve real time scheduling
- 4G (LTE Cat M1) for remotely access
- SSL/TLS support
- Two-year warranty

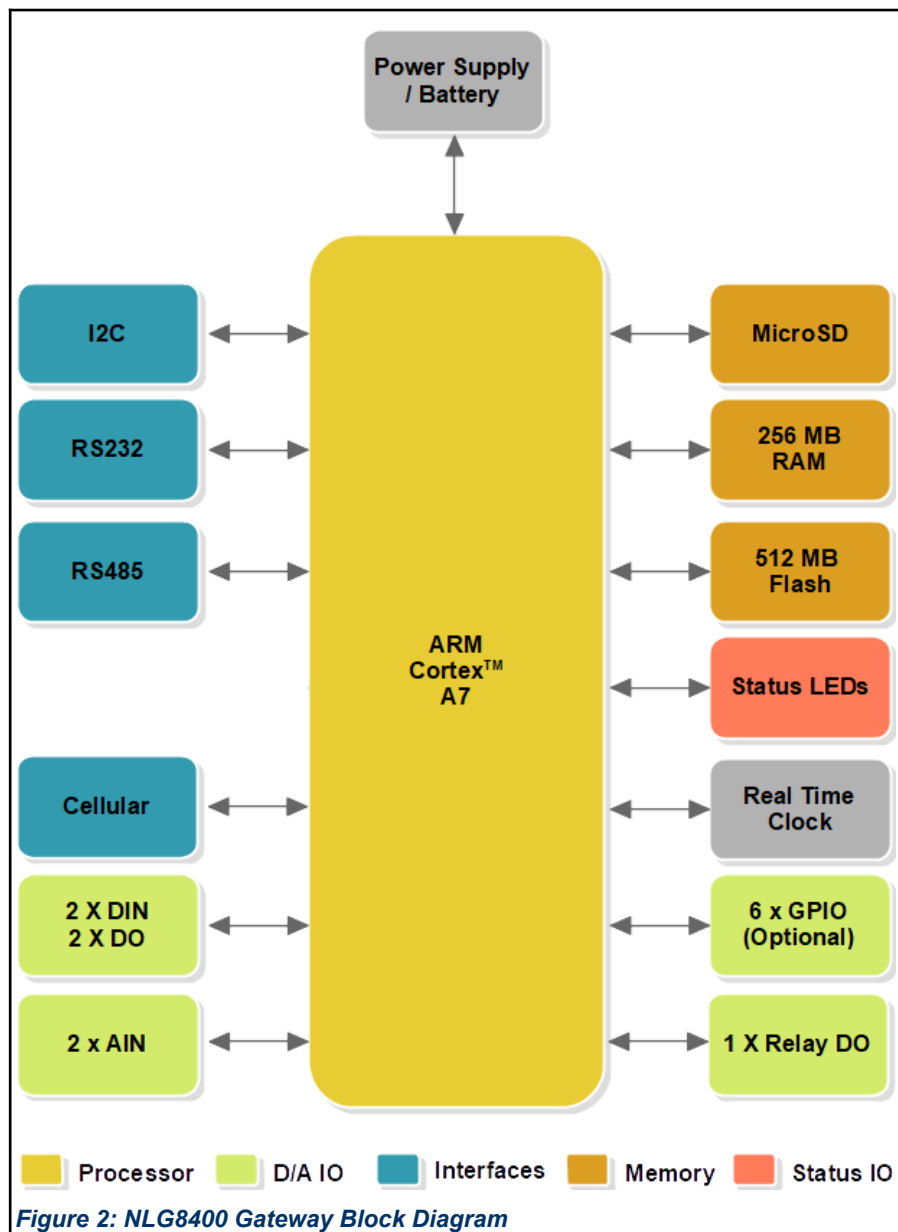
3 Application

The applications include, but are not limited to:

- Infrastructure: ITS, Utility, AMR, Lighting
- Security: Surveillance, Environment, Disaster
- Industry: Livestock, Agriculture, Heavy, Factory
- Device SCADA: PLC, IED, Generator, Sensor
- Building: Security, Energy, Waste Management, Facility, Lighting

4 Block Diagram

NebuLink Gateway NLG8400 block diagram.



5 Technical Specifications

Table – 1 Technical Specifications	
Processor	
Processor	ARM Cortex™ A7
Frequency	1.2 GHz
Storage	
External RAM	256 MB
Internal Flash	512MB Flash in which 20MB for End Application
Additional Support	Up to 32GB MicroSD Card
Software Stack	
Supported	QuecOpen™ SDK with validated Modbus RTU, Modbus TCP, DLMS Stack
Communication Interface	
Cellular	FDD-LTE:B1/B2/B3/B5/B7/B8/B20 TDD-LTE: B38/B40 HSPA/UMTS:900/850/1800/1900/2100MHz GSM/GPRS/EDGE850/900/1800/1900MHz
GPS	Optional
Serial	1 x RS232, 1 x RS485, 1 x I2C, 1 x SPI
I/O Interface	
Analogue Input	2 Channels 0-10V/4-20mA
Digital Input/Counter	Isolated 2 Channels/16 bits, Max. Input Frequency 50 Hz, Min. Pulse Width 10 ms
Digital Relay Output	Isolated 1 Channel 2A 250VAC, 220VDC Max
Digital Output	Isolated 2 Channels, 12V
General Purpose I/O (Optional)	6 Channels, 3.3VDC
General	
LEDs	3 x Status LEDs
Mechanical	
Dimensions(LxWxH) in mm	102 X 107 X 15
Electrical	
Power Supply	6 - 24 VDC @ 4A
Supported Battery Supply*	3.7/4.2 V, 2000mAH
Operating Temperature Range	-40 to +85 °C
Humidity	Relative humidity 15% to 93% non condensing
Safety	EMS Protection, ESD (IEC 61000-4-2) 4 kV Contact for each terminal, EFT (IEC 61000-4-4) ±2 kV for Power
Accessory	
Antenna	1 x Cellular, 1 x GPS (optional), 1 x Diversity (optional)
Note: * Upgradable based on the end application.	

6 I/O Specifications

Table – 2 I/O Specifications	
Digital Input	
Input Channels	2
On Voltage Level	12V
Off Voltage Level	0V
Over voltage Protection	15V, reverse stand-off
Max. Load Current	±4A Max. Continuous
Counter (Optional)	Max. Count 65536 (16bits)
Input Frequency (Max)	50 Hz
Input Pulse Width (Min)	10 ms
Digital Relay Output	
Output Channels	1
Output Type	AC / DC
Operating Voltage Range	3.3V, Voltage-Load: 0V ~ 250V
Max. Load Current	170mArms/mAdc (Continuous), ±400mA _p (Peak,10ms)
Digital Outputs	
Output Channels	2
On Voltage Level	Open Drain port
Operating Voltage Range	3.3V, Voltage-Load: 3.3V to 15VDC(Max.)
Analogue Inputs	
Input Channels	2
Voltage Range	0 – 10 V or 4 - 20 mA

7 Package & Dimension

NLG8400 Gateway package details.

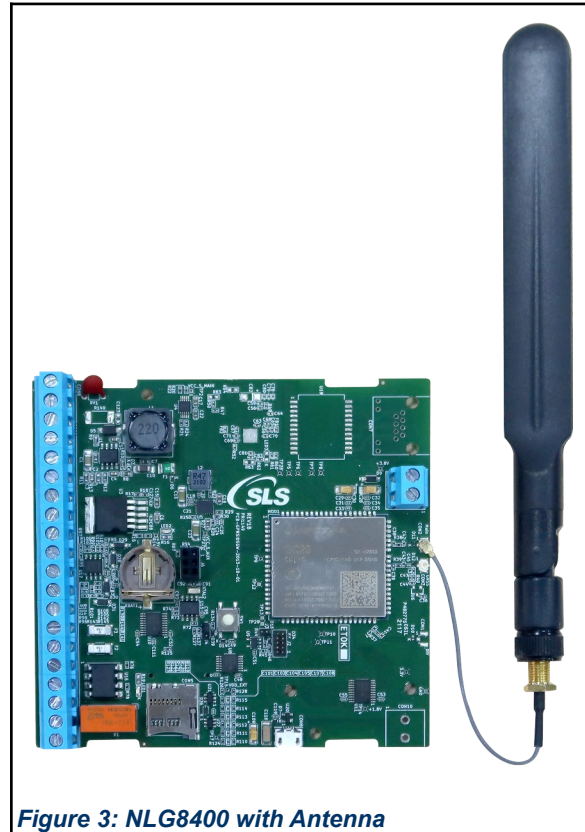


Figure 3: NLG8400 with Antenna



Figure 4: NLG8400 without Antenna

NLG8400 Gateway dimensions details.

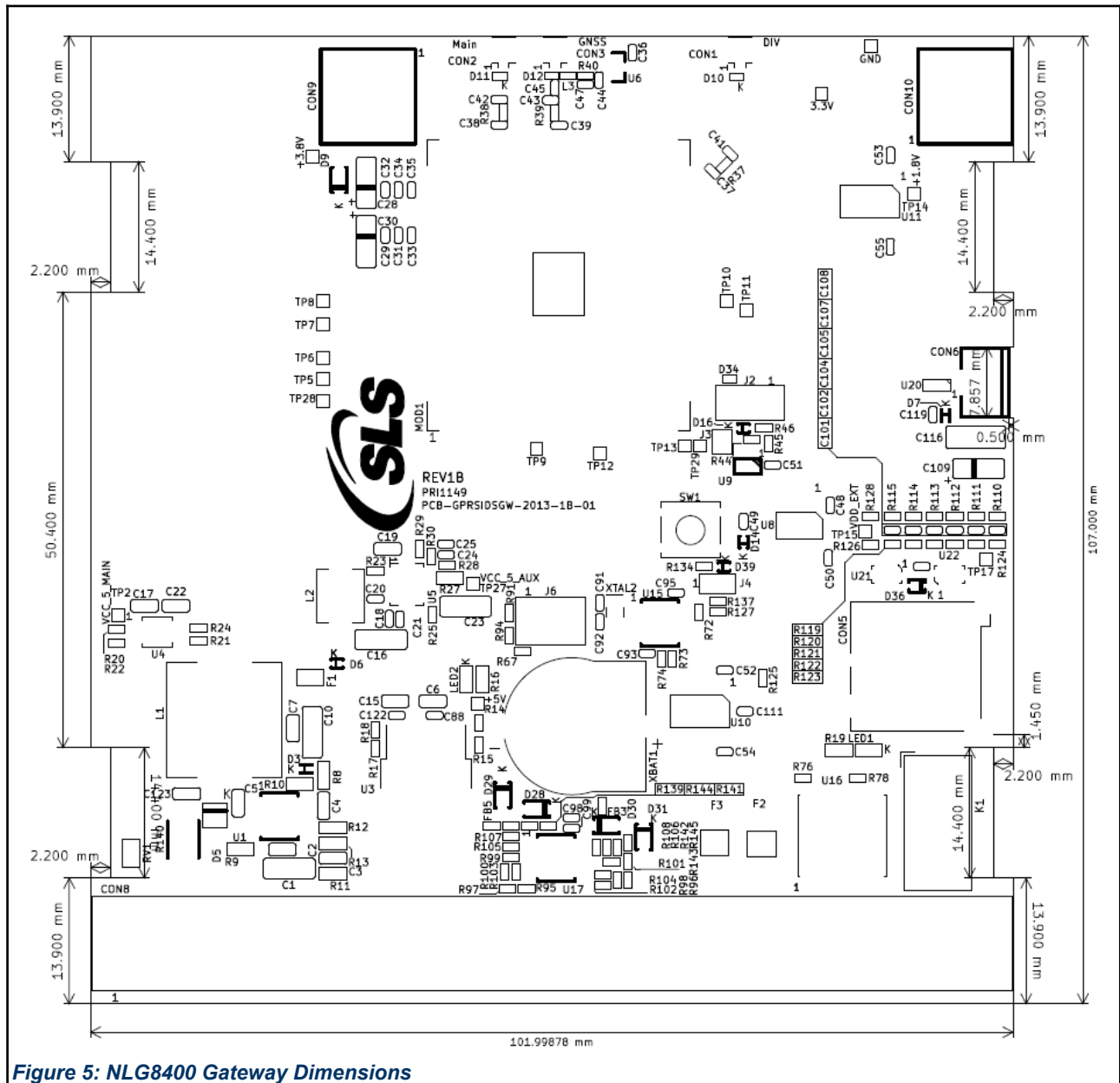


Figure 5: NLG8400 Gateway Dimensions

8 Pin Details

Table – 3 Pin Details		
Pin Number	Interface	Pin Detail
1	Power Supply	VCC
2		GND
3	Battery Supply	VIN_BATTERY
4		GND
5	ADC Input	GND
6		ADC1_IN
7		ADC2_IN
8	RS232	RS232_RTS
9		RS232_TX
10		RS232_CTS
11		RS232_RX
12	RS485	RS485-
13		RS485+
14	Digital Output	DO2
15		DO1
16	Digital Input	DIN1
17		DIN2
18		GND
19	Digital Relay Output	RELAY_NO
20		RELAY_COMM
21	---	---
22		
23		
24		
25	I2C Interface for RTC	I2C_SCL
26		I2C_SDA
27 – 40	Reserved	NC

To get started with NLG8400, follow the steps below:

1. Insert the SIM and microSD card.
2. Power on the gateway by providing power supply at pin 1 and 2 of 6-24 VDC @ 4A.
3. Follow the steps mentioned in “Setup User Guide” using QuecOpen™ SDK and validate the functionality of the gateway.

For “Setup User Guide”, contact sales@nebulae.io or visit www.nebulae.io